

Q.No 8:

DS	Session	Arrays	Question Information	Level 1	Challenge 27
Problem Description: Nirobi have given a matrix C of size $N \times M$ to Rio. Also Rio are given position of submatrix as $X1, Y1$ and $X2, Y2$ inside the matrix. Now Rio needs to find the sum of all elements inside that submatrix. Can you help Rio in completing the task assigned by Nirobi. Constraints: $1 \leq T \leq 15$ $1 \leq N, M \leq 103$ $1 \leq C[N][M] \leq 106$ $1 \leq X1, Y1, X2, Y2 \leq M$ Input Format: The first line of input contains an integer T denoting the number of test cases. The first line of each test case is n and m,n is the number of rows and m is the number of columns. The second line of each test case contains C[N][M]. The third line contains four value of X1, Y1, X2, Y2. X1, Y1 is the top left cell and X2, Y2 is the bottom right cell.					
Output Format: Print the sum of all elements inside that submatrix.					
▼ Logical Test Cases Test Case 1 INPUT (STDIN) <pre>1 3 3 91 8 71 14 2 11 6 51 3 1 2 1 3</pre> EXPECTED OUTPUT 79 Test Case 2 INPUT (STDIN) <pre>1 3 3 9 18 7 14 2 11 6 51 3 1 2 1 3</pre> EXPECTED OUTPUT 25					

Code:

```
#include <stdio.h>
int main()
{
    int T;
    scanf("%d", &T);
    while(T--)
    {
        int n, m;
        scanf("%d %d", &n, &m);

        int c[n][m];

        int i, j;
```

```

for(i = 0; i < n; i++)
{
for(j = 0; j < m; j++)
{
scanf("%d", &c[i][j]);
}
}
int x1, y1, x2, y2;
scanf("%d %d %d %d", &x1, &y1, &x2, &y2);

int sum = 0;

for(i = x1 - 1; i <= x2 - 1; i++)
{
for(j = y1 - 1; j <= y2 - 1; j++)
{
sum = sum + c[i][j];
}
}

printf("output: %d\n", sum);
}

return 0;
}

```

Output:

Test_case 1:

```

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1
3 3
91 8 71 14 2 11 6 51 3
1 2 1 3
output: 79

```

Test_case 2:

```

1
3 3
9 18 7 14 2 11 6 51 3
1 2 1 3
output: 25

```